

Experiment - 8

Name : Sharon Elango

Batch : 3

Roll No : 62

#Code for LoginForm.js :

```
import React, { useState } from 'react';

const LoginForm = () => {
  const [formData, setFormData] = useState({ username: "", email: "", password: "" });
  const [errors, setErrors] = useState({});

  const handleChange = (e) => {
    const { name, value } = e.target;
    setFormData({ ...formData, [name]: value });
  };

  const validate = () => {
    let tempErrors = {};
    if (!formData.username) tempErrors.username = "Username is required";
    if (!formData.email.includes("@")) tempErrors.email = "Invalid email format";
    if (formData.password.length < 6) tempErrors.password = "Min 6 characters required";
    setErrors(tempErrors);
    return Object.keys(tempErrors).length === 0;
  };

  const handleSubmit = (e) => {
    e.preventDefault();
    if (validate()) alert("Form Submitted!");
  };

  return (
    <div style={{ padding: '20px', border: '1px solid #ddd', margin: '10px' }}>
      <h2>Part A: Login Form</h2>
      <form onSubmit={handleSubmit}>
        <input name="username" placeholder="Username" onChange={handleChange}
        /><br/>
        {errors.username && <span style={{color:'red'}}>{errors.username}</span>}<br/>
      </form>
    </div>
  );
};
```

```

    <input name="email" placeholder="Email" onChange={handleChange} /><br/>
    {errors.email && <span style={{color:'red'}}>{errors.email}</span>}<br/>

    <input type="password" name="password" placeholder="Password"
    onChange={handleChange} /><br/>
    {errors.password && <span style={{color:'red'}}>{errors.password}</span>}<br/>

    <button type="submit">Login</button>
  </form>
  <div style={{marginTop: '10px', color: '#666'}}>
    <strong>Live Data:</strong> {formData.username} | {formData.email}
  </div>
</div>
);
};

export default LoginForm;

```

#Code for Counter.js :

```

import React, { useState } from 'react';

const Counter = () => {
  const [count, setCount] = useState(0);
  const [step, setStep] = useState(1);

  return (
    <div style={{ padding: '20px', border: '1px solid #ddd', margin: '10px' }}>
      <h2>Part B: Counter App</h2>
      <p>Count: <strong style={{fontSize: '24px'}}>{count}</strong></p>

      <label>Step: </label>
      <input
        type="number"
        value={step}
        onChange={(e) => setStep(parseInt(e.target.value) || 0)}
      /><br/><br/>
    </div>
  );
};

```

```

    <button onClick={() => setCount(count + step)}>Increase</button>
    <button onClick={() => setCount(count - step)}>Decrease</button>
    <button onClick={() => setCount(0)} style={{marginLeft: '5px'}}>Reset</button>
  </div>
);
};

export default Counter;

```

#App.jsx:

```

import React from 'react';
import LoginForm from './LoginForm';
import Counter from './Counter';

function App() {
  return (
    <div style={{ maxWidth: '600px', margin: '0 auto', fontFamily: 'Arial' }}>
      <h1 style={{ textAlign: 'center' }}>Experiment 8: State Management</h1>
      <hr />

      {/* Rendering Part A */}
      <LoginForm />

      <hr />

      {/* Rendering Part B */}
      <Counter />

    </div>
  );
}

export default App;

```

#Output :

Experiment 8: State Management

Part A: Login Form

Sharon Elango
sharonelango10@student.sfi

Submit

Live Preview:

Username: Sharon Elango

Email: sharonelango10@student.sfit.ac.in

Part B: Counter App

Step Value:

5

Observe: Clicking buttons updates state and triggers re-render without page reload.